

Phase Two: Trefoil Primitive-Cycle

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Abstract

This note fixes the *trefoil primitive-cycle* invariant used in Phase 2 of the Swirl–String Theory (SST) trefoil program. Using the ideal trefoil source `ideal.txt` (topology 3:1:1, 183 modes, $k_{\max} = 250$) and the robustness export bundle in the SSTcore repository, we define the primitive closure deviation

$$\zeta_{\text{trefoil}} \equiv \frac{a_{\text{nc}}}{r_c} - 1$$

from the no-contact radius a_{nc} and the core radius r_c . The numerical inputs A_K , A_{ratio} , a_{nc}/r_c , and the provisional stationary ratio a_*/r_c are taken from `final_best_estimate.csv`. We then introduce the associated primitive logarithmic cycle length

$$\ell_{\text{trefoil}} \equiv \log\left(\frac{a_{\text{nc}}}{r_c}\right) = \log(1 + \zeta_{\text{trefoil}}),$$

and formulate a minimal *primitive-cycle zeta ansatz* for the trefoil sector. At this stage, the primitive invariant is data-driven, while the zeta construction is a formal Phase 2.1 ansatz whose completion factor remains to be fixed in later phases by reflection-symmetry matching and comparison with the trefoil boundary transform.

Keywords: trefoil, primitive cycle, Biot–Savart, no-contact radius, Swirl–String Theory

1 Context

In the SST project, an ideal trefoil is generated from an “ideal source” identified in the export files as `ideal.txt` with topology 3:1:1, 183 modes, and $k_{\max} = 250$. The trefoil closure is analysed numerically using multiple mesh sizes and fitting procedures. The relevant outputs are stored under

```
SST_Dashboard/trefoil_closure/exports/  
sst_ideal_trefoil_robustness_outputs_v3_2nd/
```

in the SSTcore repository.

The file `final_best_estimate.csv` summarises the best estimates across the robustness runs. It records the Biot–Savart coupling constant A_K , the coupling ratio A_{ratio} , the no-contact radius ratio a_{nc}/r_c , and a provisional stationary radius ratio a_*/r_c . In the current export family, A_K and A_{ratio} are treated as the cleanest current estimates, while a_*/r_c remains provisional. The present note extracts the primitive-cycle invariant from these *v3_2nd* best-estimate exports only; any later comparison to the trefoil boundary amplitude $\Phi_{3_1}(t)$ belongs to a subsequent phase.

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Table 1: Best-estimate quantities from `final_best_estimate.csv` (systematic half-ranges as exported).

Quantity	Definition	Best estimate $\pm$ systematic half-range
Biot–Savart coupling constant for the trefoil	A_K	$0.079\,640\,93 \pm 0.006\,873\,76$
Ratio of A_K to its theoretical reference	$A_{\text{ratio}} = \frac{A_K}{1/(4\pi)}$	$1.000\,797\,46 \pm 0.086\,378\,27$
Ratio of the no-contact radius a_{nc} to the core radius r_c	a_{nc}/r_c	$1.000\,398\,65 \pm 0.041\,772\,83$
Stationary radius ratio from the full Biot–Savart model (provisional)	a_*/r_c	$1.006\,744\,74 \pm 0.045\,578\,86$

2 Definition of the trefoil primitive-cycle invariant

In the present Phase 2 setting, the *trefoil primitive-cycle invariant* is defined from the no-contact closure branch of the 3_1 trefoil sector. It measures the fractional enlargement of the loop radius required to close without self-contact:

$$\zeta_{\text{trefoil}} \equiv \frac{a_{\text{nc}}}{r_c} - 1. \quad (1)$$

If $\zeta_{\text{trefoil}} = 0$, the loop would close at the core radius itself. A positive value of ζ_{trefoil} indicates that a small radial expansion is required to avoid self-intersection.

3 Equivalent empirical forms

The exported data also include the coupling ratio

$$A_{\text{ratio}} \equiv \frac{A_K}{1/(4\pi)}, \quad \frac{1}{4\pi} \approx 0.079\,577\,47, \quad (2)$$

where $1/(4\pi)$ is the theoretical single-loop coupling constant.

For the current robustness-export family, the data suggest the empirical relation

$$\zeta_{\text{trefoil}} \approx \frac{A_{\text{ratio}} - 1}{2}. \quad (3)$$

Substituting the current best estimates,

$$A_{\text{ratio}} - 1 = 0.000\,797\,46, \quad \frac{a_{\text{nc}}}{r_c} - 1 = 0.000\,398\,65,$$

gives

$$\frac{A_{\text{ratio}} - 1}{2} \approx \frac{0.000\,797\,46}{2} \approx 0.000\,398\,73 \approx \frac{a_{\text{nc}}}{r_c} - 1.$$

Thus, within the present export family,

$$\zeta_{\text{trefoil}} \approx \frac{A_K - 1/(4\pi)}{2(1/(4\pi))}. \quad (4)$$

This relation should be understood as empirical rather than exact; its stability under future branch-selection refinements remains to be checked.

4 Numerical value and uncertainty

Using the best-estimate values from `final_best_estimate.csv`, the primitive closure deviation is

$$\zeta_{\text{trefoil}} = \frac{a_{\text{nc}}}{r_c} - 1 \approx 3.99 \times 10^{-4}. \quad (5)$$

Because the exported half-range on a_{nc}/r_c is large,

$$\frac{a_{\text{nc}}}{r_c} = 1.000\,398\,65 \pm 0.041\,772\,83,$$

the uncertainty on ζ_{trefoil} is dominated by model-dependent systematics rather than statistical noise. Accordingly, the current value of ζ_{trefoil} should be treated as a preliminary central estimate rather than as a final precision invariant.

5 Step 2.1: primitive logarithmic cycle length

From the primitive closure deviation, define the primitive logarithmic cycle length

$$\ell_{\text{trefoil}} \equiv \log\left(\frac{a_{\text{nc}}}{r_c}\right) = \log(1 + \zeta_{\text{trefoil}}). \quad (6)$$

For $\zeta_{\text{trefoil}} \ll 1$, one has the expansion

$$\ell_{\text{trefoil}} = \zeta_{\text{trefoil}} - \frac{1}{2}\zeta_{\text{trefoil}}^2 + O(\zeta_{\text{trefoil}}^3),$$

so numerically ℓ_{trefoil} is extremely close to ζ_{trefoil} at the current level of precision.

6 Step 2.1 ansatz: minimal trefoil primitive-cycle zeta

Let $\sigma = \pm 1$ label the two internal handed circulation–torsion branches of the trefoil closure sector, and let

$$\theta_{3_1}^{(\sigma)} \in \mathbb{R}$$

denote a primitive trefoil phase associated with the corresponding branch.

We then define the minimal trefoil primitive-cycle zeta ansatz by

$$Z_{3_1}^{(\sigma)}(s) = \prod_{m=1}^{\infty} \frac{1}{1 - 2\cos(m\theta_{3_1}^{(\sigma)})e^{-sm\ell_{\text{trefoil}}} + e^{-2sm\ell_{\text{trefoil}}}}, \quad \Re(s) > 0. \quad (7)$$

Here m labels repeated traversals of the same primitive trefoil cycle.

Its logarithm expands as

$$\log Z_{3_1}^{(\sigma)}(s) = 2 \sum_{m=1}^{\infty} \sum_{r=1}^{\infty} \frac{\cos(rm\theta_{3_1}^{(\sigma)})}{r} e^{-srm\ell_{\text{trefoil}}}, \quad (8)$$

where r is the standard logarithmic expansion index.

At this stage, $Z_{3_1}^{(\sigma)}(s)$ is a formal Phase 2.1 ansatz: it provides the product-side candidate associated with the primitive trefoil sector, but it is not yet tied to the trefoil boundary transform by a proven trace formula.

7 Completed primitive-cycle object

To incorporate the exact trefoil reflection symmetry introduced in the completion program, define

$$\Xi_{3_1}^{\text{pc}}(s) = G_{3_1}(s) Z_{3_1}^{(+)}(s) Z_{3_1}^{(-)}(s), \quad (9)$$

where $G_{3_1}(s)$ is a formal completion factor.

At this stage, $G_{3_1}(s)$ is not yet fixed uniquely. Its determination is deferred to the next phase, where it will be constrained by reflection-symmetry matching and comparison with the trefoil boundary transform. The target symmetry condition is

$$\Xi_{3_1}^{\text{pc}}(s) = -\Xi_{3_1}^{\text{pc}}(1-s). \quad (10)$$

The associated critical-line boundary value is

$$\Phi_{3_1}^{\text{pc}}(t) \equiv \Xi_{3_1}^{\text{pc}}\left(\frac{1}{2} + it\right). \quad (11)$$

8 Interpretation

The primitive invariant ζ_{trefoil} measures how much the trefoil closure must inflate beyond the core radius to avoid self-contact. The associated logarithmic quantity ℓ_{trefoil} is the natural primitive cycle-length variable for the product side of the trefoil duality program.

The function $Z_{3_1}^{(\sigma)}(s)$ defined above is therefore the minimal SST analogue of an Euler-product or partition-function representation built from a single primitive cycle and its repetitions. In this sense, the present construction supplies the product-side candidate for the trefoil sector, while the eventual comparison to the trefoil critical-line boundary amplitude $\Phi_{3_1}(t)$ is deferred to a later phase.

Data pointers

Raw export files (same paths on GitHub):

- https://raw.githubusercontent.com/Swirl-String-Theory/SSTcore/master/SST_Dashboard/trefoil_closure/exports/sst_ideal_trefoil_robustness_outputs_v3_2nd/final_best_estimate.csv
- https://raw.githubusercontent.com/Swirl-String-Theory/SSTcore/master/SST_Dashboard/trefoil_closure/exports/sst_ideal_trefoil_robustness_outputs_v3_2nd/final_best_estimate.txt